Automatic Question Answer Generation in Gujarati Language: A Comprehensive Literature Review

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Abstract

This document presents a comprehensive systematic literature review of automatic question generation (AQG) systems, with a specific focus on techniques applicable to developing a Gujarati language question answer generation system. The review analyzes 16 seminal papers spanning from 1976 to 2022, covering traditional rule-based approaches, machine learning techniques, and modern deep learning methods. Each paper is examined in detail, including technologies used, datasets, evaluation parameters, accuracy metrics, and training methodologies. A comprehensive summary table synthesizes all findings, and recommendations are provided for implementing a Gujarati question answer generation model based on the available dataset structure.

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1 Introduction

Automatic Question Generation (AQG) is a subfield of Natural Language Processing (NLP) that focuses on automatically generating questions from various input formats such as text, structured databases, or knowledge bases. The ability to automatically generate meaningful questions has significant applications in education, intelligent tutoring systems, chatbots, information retrieval, and assessment systems.

This literature review comprehensively analyzes 16 research papers that have contributed significantly to the field of automatic question generation. The review covers seminal works from 1976 to contemporary deep learning approaches in 2022, providing a historical perspective on the evolution of AQG technologies. Special attention is given to techniques that can be adapted for low-resource languages like Gujarati.

Research Objectives

The primary objectives of this literature review are:

1. To identify and analyze existing AQG methodologies applicable to Gujarati language
2. To understand the datasets, evaluation metrics, and training procedures used in successful AQG systems
3. To identify research gaps and challenges specific to Indic languages
4. To propose a framework for developing a Gujarati question answer generation system

Dataset Description for Gujarati AQG

The target dataset for Gujarati question answer generation consists of JSON-formatted entries with the following structure:

```
[
  {
    "context":    "                        .
                  ...",
    "source": "gujarat_history",
    "qa_pairs": [
      {
        "question_id": "q1_factual",
        "question":     "         ?",
        "answer":   "     ",
        "question_type": "factual",
        "difficulty": "easy"
      },
      ...
    ]
  }
]
```

Listing 1: Sample Gujarati QA Dataset Structure

The dataset contains multiple domains including history, wildlife, geography, literature, and general knowledge, making it suitable for developing a robust multi-domain question generation system.

2 Paper-by-Paper Review

2.1 Paper 1: "Automatic Question Paper Generation using Bloom's Taxonomy Algorithm"

Authors: Kanchan Babaji Dhomse, Dr. Ishan Ranjan
Publication: Research Scholars, SGVU, Rajasthan, India (2021)

What the Paper Says

This paper proposes an automatic question paper generation system using Bloom's Taxonomy and Machine Learning techniques. The authors argue that manual question paper creation is time-consuming and lacks standardization. Their web-based system generates domain-wise questions from datasets using ML and NLP, aiming to reduce human effort and prevent question repetition.

Technology/ML Models Used

- Bloom's Taxonomy Algorithm (for cognitive level classification)
- Colony Algorithm (optimization)
- MapReduce techniques
- Machine Learning with NLP
- Data Mining

Dataset Used

- Custom dataset of questions categorized by domain
- Not publicly available (institutional database)

Evaluation Parameters

- Accuracy comparison across years (2016-2019)
- Technologies compared: ML&NLP (95%), Artificial Intelligence (89%), Ontology (86%)

Accuracy

- ML & NLP with Data Mining: 95% (2019)
- Artificial Intelligence: 89% (2019)
- Ontology: 86% (2019)

Publication Venue

Research Scholars Publication, SGVU University, Rajasthan

Key Contributions

- Proposes hybrid model combining ML and NLP
- Addresses question repetition and duplication issues
- Web-based automated system with email integration

2.2 Paper 2: "A Review of Automatic Question Generation in Teaching Programming"

Authors: Jawad Alshboul, Erika Baksa-Varga
Publication: University of Miskolc, Hungary (2022)

What the Paper Says

This systematic review examines automatic question generation (AQG) techniques for teaching programming languages from 2017-2022. The authors analyze 18 papers and compare pre-COVID and post-COVID approaches. They find limited focus on programming-specific question generation and identify the need for better evaluation metrics and authoring tools.

Era-wise Technology Analysis

Table 1: Era-wise Technology Analysis

Era	Technology Used	Key Characteristics
Pre-COVID (2017-2019)	Bayesian Network, ITSB Tool, ANN, Vortex Optimization	Focus on supporting students in finding answers
Post-COVID (2020-2022)	JUnit, ITA, CodeTraining, PQG Model, Dynamic Codes	Focus on supporting teachers in generating online quizzes

Detailed Steps for Model Training (ANN-based Question Generation Model - 2017)

1. **Data Collection:** Collect programming questions and code snippets
2. **Preprocessing:** Tokenize code, extract features (variable names, control structures)
3. **Feature Extraction:** Use Vortex Optimization Algorithm for feature selection
4. **Training:** Train Artificial Neural Network with backpropagation
5. **Evaluation:** Compare generated questions with human-authored questions

JUnit-based Testing Example (2020):

1. Collect student code submissions
2. Parse code to identify testable components
3. Generate test cases automatically
4. Validate through user feedback

Datasets Used

- No standard dataset; papers used custom collections
- Programming exercise repositories
- Student assignment databases

Evaluation Parameters

- User feedback (teacher/student satisfaction)
- Experimental validation
- Student performance metrics
- Code coverage (for JUnit tests)

Accuracy

- Pre-COVID ANN models: 70-80% (approximate)
- Post-COVID tools: 85-90% based on user ratings

Publication Venue

University of Miskolc, Hungary

Key Contributions

- Comparative analysis of pre/post COVID approaches
- Identifies shift from student-support to teacher-support tools
- Highlights need for programming-specific AQG

2.3 Paper 3: "A Review on Question Generation from Natural Language Text"

Authors: Ruqing Zhang, Jiafeng Guo, Lu Chen, Yixing Fan, Xueqi Cheng
Publication: ACM Transactions on Information Systems (2021)
Citations: 51 | **Downloads:** 6311

What the Paper Says

This comprehensive survey provides a taxonomy of question generation tasks from three perspectives: input context text type, target answer type, and generated question type. The authors review traditional rule-based methods and advanced neural network-based methods, analyzing underlying design principles and training strategies.

Technology Classification

Rule-Based Methods (Pre-2017):

- Template-based approaches
- Syntax-based approaches (using Tregex, Tsurgeon)
- Semantic-based approaches (SRL, Minimal Recursion Semantics)

Neural Network-Based Methods (2017-2021):

Table 2: Neural Network-Based Methods Classification

Model Type	Architecture	Key Features
Seq2Seq	RNN/LSTM with attention	Answer position features, BIO tagging
Seq2Seq with Answer Span	Encoder-decoder	Binary answer indicators, relative distance
Seq2Seq with Abstractive Answer	Dual encoders	Answer masking, separate encoding
Pre-trained Models	BERT, UNILM, ERNIE-GEN, ProphetNet	Fine-tuned for QG tasks
Graph-based	GCN, GGNN	Dependency trees, entity graphs
Generative Models	VAE, GAN	Diverse question generation

Detailed Training Steps (Seq2Seq Model Example)

Step 1: Data Preparation

```
Input: Passage P, Answer span A
Example: "Warsaw was given authority...", answer: "Warsaw"
Tokenize and create vocabulary
Add answer position features (BIO tags: B-answer, I-answer, O)
```

Step 2: Encoder Architecture

```
Input: [CLS] Warsaw was given authority [SEP]
Word embeddings + Position embeddings + Answer tag embeddings
Feed into Bi-LSTM
Output: Hidden states h1, h2, ..., hn
```

Step 3: Attention Mechanism

$$\text{Attention weights} = \text{softmax}(\text{decoder_state} \cdot \text{encoder_hidden_states})$$

$$\text{Context vector} = \sum(\text{attention_weights} \times \text{encoder_hidden_states})$$

Step 4: Decoder

```
Initialize with encoder final state
At each step: predict next word using previous word + context vector
Use copy mechanism for out-of-vocabulary words
```

Step 5: Training Objective

$$\text{Loss} = -\sum \log P(y_t | y_{<t}, x) \quad \text{(Negative log-likelihood)}$$

- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 32
- Epochs: 20-30

Datasets Used

Dataset	Size	Type	Link
SQuAD 1.1	100K QA pairs	Answer span	https://rajpurkar.github.io/SQuAD-explorer/
SQuAD 2.0	150K (50K unanswerable)	Answer span	https://rajpurkar.github.io/SQuAD-explorer/
NewsQA	100K QA pairs	Answer span	https://www.microsoft.com/en-us/research/project/newsqa-dataset/
SearchQA	140K QA pairs	Answer span	https://github.com/nyu-dl/dl4ir-searchQA
HotpotQA	113K QA pairs	Multi-hop reasoning	https://hotpotqa.github.io/
MS MARCO	1M questions	Abstractive answer	https://microsoft.github.io/msmarco/

NarrativeQA	46K QA pairs	Abstractive answer	https://github.com/deepmind/narrativeqa
CoQA	127K conversational QA	Sequential questions	https://stanfordnlp.github.io/coqa/
QuAC	100K QA dialogues	Sequential questions	https://quac.ai/
RACE	97K MCQs	Multiple choice	http://www.cs.cmu.edu/~glai1/data/race/
LearningQ	230K document-question	Educational	https://github.com/persist-gt/learningq

Evaluation Parameters

- **BLEU (Bilingual Evaluation Understudy):** n-gram precision (BLEU-1,2,3,4)
- **ROUGE-L:** Recall-oriented, longest common subsequence
- **METEOR:** Recall-oriented with synonym matching
- **Human evaluation:** Fluency, relevance, grammaticality

Accuracy Results (SQuAD dataset)

Table 4: Accuracy Results on SQuAD Dataset

Model	**BLEU-4**	**METEOR**	**ROUGE-L**
Seq2Seq (Du et al. 2017)	12.28	16.62	39.75
M&N 2015	15.32	19.29	43.91
GE_ROUGE+QSS+ANSS (2019)	16.48	20.21	44.11
CS2S-VR-ACS (2020)	22.05	25.11	53.25
ERNIE-GEN (2020)	25.40	26.92	52.84
ProphetNet (2020)	25.01	26.83	52.57

Publication Venue

ACM Transactions on Information Systems (TOIS), Volume 40, Article 14

Key Contributions

- Comprehensive taxonomy of QG tasks
- Detailed analysis of 100+ papers
- Empirical comparison on benchmark datasets
- Identifies trending topics: diverse QG, pre-training, higher cognitive levels

2.4 Paper 4: "A Systematic Review of Automatic Question Generation for Educational Purposes"

Authors: Ghader Kurdi, Jared Leo, Bijan Parsia, Uli Sattler, Salam Al-Emari
Publication: International Journal of Artificial Intelligence in Education (2020)

What the Paper Says

This systematic review extends previous work (Alsubait 2015) and covers 93 papers from 2015-2019 on educational AQG. The authors analyze publication trends, research groups, generation methods, evaluation techniques, and identify gaps in difficulty control, feedback generation, and verbalization.

Publication Statistics

- 93 papers included
- 45% increase from previous review (81 papers over 45 years → 93 papers over 4 years)
- 44 conference papers, 32 journal articles, 17 workshop papers
- 358 researchers identified (168 from previous review + 205 new, 15 overlap)

Era-wise Technology Analysis

Table 5: Era-wise Technology Analysis

Time Period	Dominant Technology	Key Characteristics
Pre-2015	Rule-based, Template-based	Manual rule creation, domain-specific
2015-2019	Semantic-based, ML-based	WordNet, ontologies, statistical methods
2015-2019	Neural methods (emerging)	Seq2Seq, attention mechanisms

Generation Methods Classification

By Understanding Level:

1. **Syntactic-based** (10 papers): POS tags, parse trees
2. **Semantic-based** (60 papers): Ontologies, WordNet, similarity measures

By Transformation Procedure:

1. **Template-based** (27 papers): Predefined templates with slots
2. **Rule-based** (16 papers): Transformation rules
3. **Statistical/ML** (9 papers): Trained from data

Detailed Steps for Template-based Question Generation

Step 1: Input Processing

```
Input sentence: "The verb is the most central element in a clause."
POS tagging: [The/DT] [verb/NN] [is/VBZ] [the/DT] [most/RBS] [central/JJ] [element/NN] [in/IN] [a/DT] [clause/NN]
```

Step 2: Pattern Matching

```
Pattern: "The [KEY] is [DESCRIPTION]"
Match found: KEY = "verb", DESCRIPTION = "the most central element in a clause"
```

Step 3: Template Selection

```
Question template: "Which part of speech [DESCRIPTION]?"
Hypernym mapping: "verb" -> hypernym = "part of speech" (using WordNet)
```

Step 4: Question Generation

```
Generated question: "Which part of speech is the most central element in a clause?"
Answer: "verb"
```

Step 5: Distractor Generation

```
Using WordNet: synonyms/related terms of "verb"
Distractors: "noun", "adjective", "adverb"
```

Datasets Used

Table 6: Datasets Used in Educational AQG

Dataset	Size	Purpose	Link/Availability
SQuAD	150K	Reading comprehension	Public
NewsQA	120K	Reading comprehension	Public
SciQ	13.7K	Science MCQs	Public
MCQL	7.1K	MCQs from web	Web crawl
QGSTEC	570	QG shared task	http://www.cs.memphis.edu/~vrus2/QGSTEC/
Medical CBQ	435	Medical case-based	Custom (EMMeT ontology)

Evaluation Parameters

Quality Criteria Used

Accuracy Results (Sample)

Publication Venue

International Journal of Artificial Intelligence in Education (2020) 30:121-204

Table 7: Evaluation Types and Metrics

Evaluation Type	**Number of Studies**	**Metrics Used**
Expert review	21	Acceptability, grammaticality, relevance
Human comparison	15	Overlap, BLEU, ROUGE, METEOR
Crowdsourcing	10	Quality ratings
Student testing	19	Difficulty, discrimination

Table 8: Quality Criteria Frequency

Criteria	**Number of Studies**
Acceptability	17
Grammatical correctness	14
Semantic ambiguity	11
Educational usefulness	10
Difficulty	19
Distractor quality	16

Key Contributions

- Comprehensive analysis of 93 papers
- Identifies research gaps: difficulty control, feedback generation, verbalization
- Provides publication venues and active research groups
- Quality assessment framework for AQG evaluation

Table 9: Sample Accuracy Results

Study	Domain	Accuracy
Liu et al. (2017)	Generic	66.2-69.4% correct sentences
Aldabe et al. (2006)	Language	>80% quality
Agarwal & Mannem (2011)	Biology	91.66% sentence selection, 94.16% key selection
Majumder & Saha (2015)	Generic	93.21% informative sentences, 83.03% key selection

2.5 Paper 5: "Automatic Generation and Ranking of Questions for Critical Review"

Authors: Ming Liu, Rafael A. Calvo, Vasile Rus
Publication: Journal of Educational Technology & Society (2014)

What the Paper Says

This paper presents an automatic question generation system for academic writing support that generates trigger questions from citations. The system uses an overgeneration-and-ranking approach with learning-to-rank algorithms (RankSVM and Logistic Regression) to improve question quality. The generated questions help students reflect on their literature reviews.

Technology/ML Models Used

- **RankSVM** (Pairwise ranking approach)
- **Logistic Regression** (Pointwise ranking approach)
- Named Entity Recognition (LBJ NER tagger)
- Tregex for syntactic pattern matching
- Naive Bayes classifier for citation classification

Dataset Used

- **Training set:** 504 questions from 45 academic papers (ACL and IJCAI conferences)
- **Testing set:** 489 questions from 33 literature reviews by engineering research students
- Not publicly available (academic paper corpus)

Detailed Steps for Model Training

Step 1: Preprocessing

```
Input paper -> Extract citation sentences using pattern matching
Example citation: "Cannon (1927) challenged this view mentioning that
    physiological changes were not sufficient to discriminate emotions."
```

Step 2: Sentence Simplification

```
Complex sentence -> Split into simpler sentences using Tregex
Simplified: "Cannon challenged this view mentioning that physiological
    changes were not sufficient to discriminate emotions."
```

Step 3: Passive-to-Active Transformation

```
Passive detection -> Apply transformation rules
Example: "It is argued that..." -> "Peter argued that..."
```

Step 4: Citation Classification (Naive Bayes) Features used:

- Cue phrases (17 features)
- Sentiment features (SENTIWORDNET)
- Negation features
- Syntactic features

Classify into: Opinion, Result, Aim, System, Method, Application

Step 5: Question Generation (Template-based)

Table 10: Question Generation Templates by Category

Rule/Category	**Question Templates**
Rule 1 (Opinion)	”Why + subject_auxiliary_inversion?” ”What evidence is provided by +subject+ to prove the opinion?” ”Does any other scholar agree or disagree with +subject+?”
Rule 2 (Aim)	”Why does +subject+ conduct this study?” ”What is the research question?”
Rule 3 (Result)	”subject_auxiliary_inversion?” ”Is the analysis accurate?”

Generated example:

```
"Why did Cannon challenge this view?"
"What evidence is provided by Cannon to prove the opinion?"
```

Step 6: Feature Extraction for Ranking (11 features)

1. Num. of Named Entities: Count of author names
2. IsPronounResolved: Boolean
3. IsPassiveVoice: Boolean
4. ContainsAnswer: Boolean (cue phrases like “due to”)
5. Negation: Boolean
6. PredicationConfidence: Numeric (classifier confidence)
7. ReportingVerb: Boolean
8. NameAppearInBoth: Boolean
9. NameAppearInSubject: Boolean

10. Num. of Clauses: Numeric
11. Length: Numeric

Step 7: Ranking Model Training
RankSVM Training:

$$\text{Objective: Learn weight vector w such that } f(q_i) = w \cdot q_i$$

$$\text{For question pairs } (q_i, q_j): \text{ if } q_i \text{ preferred over } q_j, \text{ then } w \cdot q_i > w \cdot q_j$$

$$\text{Minimize: } \frac{1}{2}\|w\|^2 + C \sum \xi_{ij}$$

$$\text{Subject to: } w \cdot q_i \geq w \cdot q_j + 1 - \xi_{ij}, \quad \xi_{ij} \geq 0$$

Logistic Regression Training:

$$P(\text{acceptable}|q) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_{11} x_{11})}}$$

Train using binary labels (acceptable/unacceptable)

Evaluation Parameters

- **Accuracy at top K:** Percentage acceptable in top K ranked questions
- **Cohen's Kappa:** Inter-annotator agreement (0.65)
- **5 Quality Measures** (Likert 1-5 scale):
  - QM1: Correctly written
  - QM2: Clear
  - QM3: Appropriate to context
  - QM4: Makes me reflect
  - QM5: Useful question

Accuracy Results

Table 11: Ranking Model Performance

Model	Top 25% Accuracy	Top 50% Accuracy
RankSVM (All features)	75.8%	71.3%
RankSVM (Semantic only)	73.3%	71.3%
RankSVM (Basic only)	70.8%	69.2%
Logistic Regression	74.2%	71.3%
Baseline	65.7%	65.6%

Question Quality Comparison:

Publication Venue

Journal of Educational Technology & Society, 17(2), 333-346 (2014)

Table 12: Question Quality Comparison

Question Producer	**QM1**	**QM2**	**QM3**	**QM4**	**QM5**	**Average**
AQG System	4.26	4.18	4.06	3.99	4.02	4.10
Supervisor	4.57	4.53	4.45	4.26	4.21	4.40
Peer	3.93	3.96	3.77	3.62	3.56	3.77
Generic	4.04	3.92	3.65	3.49	3.57	3.73

Key Contributions

- Novel overgeneration-and-ranking approach for QG
- 11-feature ranking model for question acceptability
- Comparison of RankSVM vs Logistic Regression
- Analysis of human question types (Association, Explanation, Prediction)

2.6 Paper 6: "Automatic Multiple Choice Question Generation From Text: A Survey"

Authors: Dhawaleswar Rao CH, Sujan Kumar Saha
Publication: IEEE Transactions on Learning Technologies (2020)

What the Paper Says

This systematic survey reviews 86 papers on automatic MCQ generation from text from 1997-2018. The authors establish a generic 6-phase workflow for MCQ generation and analyze techniques for each phase. They identify research gaps including handling multi-line facts, complex distractors, and standard evaluation metrics.

6-Phase Workflow for MCQ Generation

Phase 1: Pre-processing

- Text Normalization
- Structural Analysis
- Sentence Simplification
- Lexical Analysis (stemming)
- Statistical Analysis (TF-IDF, n-gram)
- Syntactic Analysis (POS tagging, NER, parsing)
- Coreference Resolution
- Word Sense Disambiguation

Phase 2: Sentence Selection

- Sentence length
- Occurrence of particular words
- Sentence position
- Presence of named entities
- Topic modeling
- Similarity to document title
- Pattern matching with question corpora

Phase 3: Key Selection

- Noun phrases (most common: 50+ papers)
- Term frequency
- Named entity recognition

- Semantic role labeling
- Machine learning classifiers
- Topic modeling

Phase 4: Question Formation

- Wh-word selection (what, who, where, when)
- Subject-Verb-Object analysis
- Dependency-based patterns
- Syntactic transformation
- Discourse connectives
- Semantic role-based templates

Phase 5: Distractor Generation

- Parts-of-speech information
- Word frequency
- WordNet (synonyms, hypernyms)
- Domain ontology
- Distributional similarity
- Latent Semantic Analysis
- Word2vec embeddings

Phase 6: Post-processing

- Question post-editing
- Question filtering
- Question ranking

Detailed Example of MCQ Generation

Input Sentence:

```
"The nucleus contains positively charged particles called protons and
    electrically neutral particles called neutrons."
```

Step 1: Sentence Selection

- Informative sentence with factual content
- Contains scientific terms

Step 2: Key Selection

- Identify candidate: “neutrons” (noun phrase)
- Check: factual, testable concept

Step 3: Question Formation

```
Using semantic pattern: "A1 contains A2 called A3"
Pattern detected: "The nucleus [A1] contains [V] positively charged
    particles called protons [A2] and electrically neutral particles called
    neutrons [A3]"
Question type: "What is another term for [A3]?"
Generated: "What is another term for electrically neutral particles?"
```

Step 4: Distractor Generation

```
Using WordNet/ontology:
- Synonyms/hyponyms of "neutrons": "protons", "electrons", "atoms"
- Similar named entities: other subatomic particles
Final distractors: "protons", "electrons", "atoms"
```

Final MCQ:

```
What is another term for electrically neutral particles?
a) protons
b) electrons
c) atoms
d) neutrons (correct)
```

Datasets Used

- SQuAD (100K) - https://rajpurkar.github.io/SQuAD-explorer/
- NewsQA (120K) - https://www.microsoft.com/en-us/research/project/newsqa-dataset/
- SciQ (13.7K) - https://allenai.org/data/sciq
- MCQL (7.1K) - Custom
- QGSTEC (570) - http://www.cs.memphis.edu/~vrus2/QGSTEC/

Evaluation Parameters

Stem and Key Evaluation:

- Well-formedness
- Sentence length
- Sentence simplicity
- Difficulty level
- Domain relevance
- Grammatical correctness
- Over-informativeness/under-informativeness

Distractor Evaluation:

- Plausibility
- Semantic closeness to key
- Functionality (chosen by examinees)
- Homogeneity
- Readability

Accuracy Results

Table 13: MCQ Generation Accuracy Results

Study	**Domain**	**Accuracy**
Liu et al. (2005)	Generic	66.2-69.4% correct sentences
Aldabe et al. (2006)	Language	>80% quality
Agarwal & Mannem (2011)	Biology	91.66% sentence selection, 94.16% key selection
Bhatia et al. (2013)	Wikipedia	88% distractor accuracy, 79.4% key accuracy
Majumder & Saha (2015)	Generic	93.21% informative sentences, 83.03% key selection
Kumar et al. (2015)	Generic	94% sentences, 87% gaps, 60% distractors

Publication Venue

IEEE Transactions on Learning Technologies, Vol. 13, No. 1, January-March 2020

Key Contributions

- First comprehensive survey on MCQ generation
- 6-phase generic workflow
- Analysis of 86 papers
- Identifies research gaps: complex distractors, multi-line facts, standard evaluation

2.7 Paper 7: "Automatic question generation approaches and evaluation techniques"

Authors: Manisha Divate, Ambuja Salgaonkar
Publication: Current Science, Vol. 113, No. 9, November 2017

What the Paper Says

This review paper analyzes automatic question generation systems, their approaches, and evaluation techniques. It categorizes question types (gap-fill, MCQ, factoid-based, deep learning questions) and describes the AQG process including sentence simplification, answer phrase selection, sentence transformation, and ranking.

Question Type Classification

Table 14: Question Type Classification

Question Type	**Description**	**Example**
Gap-fill	Remove word/phrase from sentence	"The ____ is the most central element in a clause."
MCQ	Multiple choice with distractors	"Which part of speech is the most central element?"
Factoid-based	Wh-questions	"When did Tilak obtain Bachelor of Arts?"
Deep learning	Higher-order thinking	"Why did Cannon challenge this view?"

Technology/ML Models Used

Sentence Simplification Techniques:

- Independent clause splitting
- Appositive phrase removal
- Prepositional phrase filtering
- Relative clause handling
- Discourse marker removal
- Noun participle processing

Answer Phrase Selection:

- Noun phrases
- Prepositional phrases

- Semantic role labeling (SRL)
- Named entity recognition (NER)

Question Generation Approaches:

Table 15: Question Generation Approaches Comparison

Approach	Description	Advantages	Disadvantages
Template-based	Predefined templates with slots	Simple, effective	Limited diversity
Syntax-based	Transform parse trees	Linguistically grounded	Error propagation
Semantic-based	Use semantic roles	Deeper questions	Requires robust SRL

Ranking Models:

- Logistic Regression (pointwise)
- RankSVM (pairwise)
- Regression SVM
- J48 classifier

Detailed Example of Syntax-based Question Generation

Step 1: Input Sentence

```
Sentence: "Dhoni plays cricket."
Parse tree: (S (NP Dhoni) (VP plays (NP cricket)))
```

Step 2: Identify Answer Phrase

```
Answer phrase: "cricket" (NP object)
```

Step 3: Mark Unmovable Phrases

```
"Dhoni plays cricket" -> mark main verb
```

Step 4: Decompose Main Verb

```
"plays" -> "does" + "play"
"Dhoni does play cricket"
```

Step 5: Subject-Auxiliary Inversion

```
Yes/No question: "Does Dhoni play cricket?"
```

Step 6: Insert Wh-word

```
"What does Dhoni play?" (since answer is object)
```

Detailed Example of Semantic-based Question Generation

Step 1: Semantic Role Labeling (Illinois SRL)

```
Sentence: "Yesterday teacher taught us English in class."

Semantic roles:
- [AM-TMP] Yesterday
- [A0] teacher
- [V] taught
- [A2] us
- [AM-LOC] in class
```

Step 2: Pattern Matching

```
Pattern for "who" question: Replace A0 with "who"
Pattern for "where" question: Replace AM-LOC with "where"
```

Step 3: Question Generation

```
Who question: "Who taught students English in class yesterday?"
Where question: "Where did teacher teach students English yesterday?"
```

Feature Set for Ranking (16-dimensional)

Table 16: Ranking Feature Set

Feature Category	**Attributes**
Length	Answer phrase length, source sentence length, question length
Grammatical	Proper nouns, pronouns, adjectives, adverbs, conjunctions
Transformable	Removal of appositives, subject as answer
Semantic	Noun, verb, preposition
Negation	Not, nor, never
Vagueness	Vague noun phrases

Evaluation Parameters

Table 17: Evaluation Types

Evaluation Type	**Description**
Intrinsic evaluation	Against gold standard
Extrinsic evaluation	Impact on external task
Black-box evaluation	Speed, reliability, accuracy
Glass-box evaluation	Test algorithms, resources
Automatic evaluation	Compare with gold standard
Manual evaluation	Human judges
Bystander Turing Test	Human vs machine differentiation

Table 18: Precision, Recall, F-score Comparison

AQG System	Precision	Recall	F-score
Liu et al. (2012)	0.73	0.71	0.7
Ali et al. (2010)	0.587	0.276	0.38
Le & Pinkwart (2015)	0.796	0.276	0.41
Lindberg (2013)	0.79	-	-

Table 19: Cohen's Kappa Values

AQG System	Cohen's Kappa
Agarwal & Mannem (2011)	0.7
Zhao et al. (2011)	0.78
Liu et al. (2012)	0.57
Liu et al. (2014)	0.65
Le & Pinkwart (2015)	0.086

Accuracy Results

Precision, Recall, F-score Comparison:
Cohen's Kappa Values:
Syntactic Deficiency Analysis:

Table 20: Syntactic Deficiency Analysis (%)

Deficiency	HSAQG	Divate & Salgaonkar (2015)	McConnell (2011)	Lindberg
Ungrammatical	14%	5.17%	35.5%	85
Doesn't make sense	20.6%	4.31%	33.6%	63
Vague	19.6%	11.21%	23.4%	78
Wrong Wh-word	4.9%	30.17%	20.6%	-

Publication Venue

Current Science, Vol. 113, No. 9, 10 November 2017

Key Contributions

- Comprehensive comparison of AQG approaches
- Detailed feature set for ranking
- Analysis of evaluation techniques
- Introduces concept of Automatic Question Quality Enhancer (AQQE)

2.8 Paper 8: "Automatic Question Generation for Literature Review Writing Support"

Authors: Ming Liu, Rafael A. Calvo, Vasile Rus
Publication: International Conference on Intelligent Tutoring Systems (2010)

What the Paper Says

This paper presents an AQG system that generates trigger questions from citations in students' literature reviews to support academic writing. The system uses NLP techniques to extract citations, classify them, and generate content-related questions. A pilot study using Bystander Turing Test shows that human evaluators have difficulty distinguishing system-generated questions from human-generated ones.

Technology/ML Models Used

- Pattern matching for citation extraction (regular expressions)
- Named Entity Recognition (LBJ NER tagger)
- Pronoun resolver (simple nearest entity)
- Tregex for syntactic feature extraction
- Stanford Parser for phrase structure trees
- WordNet for verb lemma extraction
- Rule-based citation classification
- SENTIWORDNET for sentiment analysis
- Template-based question generation

Dataset Used

- 6 literature reviews from postgraduate engineering students
- 1,088 sentences including 221 citations
- 20 questions per student (5 from each source: tutor, lecturer, system, generic)
- Custom academic dataset (not publicly available)

Detailed Steps for Model Training

Step 1: Citation Extraction (Regular Expressions)

```
Pattern for textual syntactic style:
\([a-zA-Z]*\s*\d{4}\)
Pattern for textual parenthetical style:
\([a-zA-Z]*\s*\d{4}\)

Example match: "(Cannon 1927)"
```

Step 2: Syntactic Feature Extraction (Tregex)

```
Subject extraction: "NP > (S > ROOT)"
Predicate verb extraction: "/VB > (VP > (S > ROOT))"
Predicate extraction: "VP > (S > ROOT)"

Voice detection using dependency relations:
- nsubjpass indicates passive voice
```

Step 3: Citation Classification (Rule-based)
Reporting Verb Categories:

```
Opinion verbs: challenge, argue, state, believe
Aim verbs: focus, aim, attempt, seek
Result verbs: show, demonstrate, find, report
```

Tregex patterns for System/Method:

```
Method: "VP>(S>ROOT)<<,(use|apply)<<(NP<<-(method|approach))"
System: "NP > (S > ROOT) << (system|tool)"
```

Sentiment detection using SENTIWORDNET:

```
If no reporting verb match, check sentiment scores
Positive/negative scores > threshold -> Opinion category
```

Step 4: Question Generation (Template-based)

Table 21: Question Generation Rules and Templates

Rule	Pattern	Question Templates
Rule 1	Reporting Verb + Opinion	"Why + subject_auxiliary_inversion?" "What evidence is provided by +subject+?" "Does any other scholar agree or disagree?"
Rule 2	Reporting Verb + Aim	"Why does +subject+ conduct this study?" "What is the research question?" "What is +subject+'s contribution?"
Rule 3	Reporting Verb + Result	"subject_auxiliary_inversion?" "Is the analysis accurate?" "How does it relate to your research?"
Rule 4	Tregex Method	"In the study of +subject+, why...?" "Which dataset does +subject+ use?"
Rule 5	Tregex System	"In the study of +subject+, why...?" "What are the strengths and limitations?"
Rule 6	Sentiment + No Reporting Verb	Similar to Opinion templates
Rule 7	No Sentiment + No Reporting Verb	Generic templates

Evaluation Parameters

- **Citation extraction accuracy:** 65.60% (145/221 citations)
- **Semantic correctness:** 60% average (human annotators, Cohen's κ=0.61)
- **5 Quality Measures** (Likert 1-5 scale):
  - QM1: Correctly written

  - QM2: Clear
  - QM3: Appropriate to context
  - QM4: Makes me reflect
  - QM5: Useful question

- **Bystander Turing Test:** Identify source (lecturer/tutor/system)

Accuracy Results

Question Generation by Rule:

Table 22: Question Generation Accuracy by Rule

Rule	Generated	Correct	Accuracy
Rule 1	10	6	60%
Rule 2	9	5	56%
Rule 3	14	8	57%
Rule 4	7	4	57%
Rule 5	17	10	59%
Rule 6	98	56	57%
Rule 7	6	4	67%
Total	**161**	**97**	**60%**

Question Quality Comparison:

Table 23: Question Quality Comparison

Producer	QM1	QM2	QM3	QM4	QM5	Avg
AQG System	4.26	4.18	4.06	3.99	4.02	4.10
Lecturer	4.57	4.53	4.45	4.26	4.21	4.40
Tutor	3.93	3.96	3.77	3.62	3.56	3.77
Generic	4.04	3.92	3.65	3.49	3.57	3.73

Statistical Significance (t-tests):

- AQG vs Generic: Significant for QM5 (p=0.020)
- AQG vs Tutor: Not significant for QM1 and QM5
- AQG vs Lecturer: No significant differences

Bystander Turing Test Results:

Table 24: Bystander Turing Test Confusion Matrix

Real \ Response	Tutor	Lecturer	AQG
Tutor	0.7	2.7	1.6
Lecturer	0.8	1.2	3.0
AQG	1.4	1.0	2.6

F-score for AQG system: 0.43 (moderate difficulty to distinguish)

Publication Venue

International Conference on Intelligent Tutoring Systems (ITS 2010), LNCS 6094, pp. 45-54

Key Contributions

- Novel AQG for academic writing support
- Citation-based question generation
- Bystander Turing Test evaluation methodology
- Comparison with human-generated questions

2.9 Paper 9: "Automatic Question Generation from Sentences"

Authors: Husam Ali, Yllias Chali, Sadid A. Hasan
Publication: University of Lethbridge, Canada (TALN 2010)

What the Paper Says

This paper presents a target-driven automatic question generation system that takes sentences as input and generates factoid questions. The system simplifies complex sentences into elementary sentences using syntactic parsing, classifies sentences based on subject-verb-object-preposition patterns, and generates questions using interaction rules. Evaluation on TREC-2007 dataset shows promising results.

Technology/ML Models Used

- Oak system for tokenization and POS tagging
- Named Entity Recognition (Oak system, 150 NE types)
- Charniak parser for syntactic parsing
- Two-layered sentence classifier (fine + coarse)
- Rule-based question generation (90 interaction rules)

Dataset Used

- **TREC-2007 Question Answering Track dataset**
- 70 topics, 50 files per topic
- Contains questions, answers, and source files
- **Dataset link:** https://trec.nist.gov/data/qa.html (freely available for research)

Detailed Steps for Model Training

Step 1: Data Preprocessing

```
Clean HTML/XML tags from documents
Tokenize using Oak system
Extract topic number, answer, and source file from answer file
Search source files for sentences containing answer/target
```

Step 2: POS and NE Tagging (Oak system)

```
POS tags: [The/DT] [boy/NN] [went/VBD] [to/TO] [school/NN] [on/IN] [Monday/
    NNP]
NE tags: [Monday/DATE], [school/ORGANIZATION?], [boy/PERSON?]
```

Step 3: Elementary Sentence Construction (Charniak parser)

Parse tree for complex sentence:

```
(TOP (S (NP (DT The) (NN boy))
      (VP (VBD went)
          (PP (TO to)
              (NP (NN school)))
          (PP (IN on)
              (NP (NNP Monday))))
      (. .)))
```

Extract elementary sentences by combining NPs with VPs and PPs:

```
Elementary sentence 1: "The boy went to school"
Elementary sentence 2: "The boy went on Monday"
```

Step 4: Sentence Classification (Two-layered)
Coarse Classes (5):

1. Human: person names
2. Entity: animals, plants, objects
3. Location: country, city, school
4. Time: year, Monday, 9 am, last week
5. Count: 9 men, measurements

Fine-to-Coarse Mapping:

```
Fine classes from Oak (150 types) -> Map to coarse classes
Example: PERSON, DOCTOR, TEACHER -> Human
Example: CITY, COUNTRY, STATE -> Location
```

Interaction Rules (90 rules):

```
Rule format: "Human Verb Human" -> Question type: "whom and who"
Rule format: "Human Verb Entity" -> Question type: "what"
Rule format: "Human Verb Human Time" -> Question type: "when" (add when)
```

Step 5: Question Generation
For sentence: "John plays cricket in the evening"

```
Classification: Human (John) + Verb (plays) + Entity (cricket) + Time (
    evening)
Rule match: "Human Verb Entity Time"
Generated question types: what, when
```

What question: "What does John play in the evening?" **When question:** "When does John play cricket?"

Evaluation Parameters

$$\text{Recall} = \frac{Q_g \cap Q_a}{Q_a} \quad (\text{generated} \cap \text{actual/actual})$$

$$\text{Precision} = \frac{Q_g \cap Q_r}{Q_r} \quad (\text{generated} \cap \text{relevant/relevant})$$

Relevant questions exclude grammatical errors

Table 25: Recall by Question Type

Question Type	Generated (Q_g)	Actual (Q_a)	Recall
What	75	520	0.135
Which	3	10	0.300
Who/Whom	20	40	0.550
How many/much	3	13	0.231
Where	4	8	0.500
When	1	2	0.500
Overall	**29**	**105**	**0.276**

Table 26: Precision by Question Type

Question Type	Generated (Q_g)	Relevant (Q_r)	Precision
What	105	43	0.410
Which	57	23	0.404
Who/Whom	144	106	0.736
How many/much	43	17	0.395
Where	117	89	0.761
When	71	37	0.521
Overall	**537**	**315**	**0.587**

Accuracy Results

Recall Evaluation (20 topics with generated questions):
Precision Evaluation (5 topics):

Publication Venue

TALN 2010 (Traitement Automatique des Langues Naturelles), Montreal, Canada

Key Contributions

- Target-driven QG approach
- Two-layered sentence classification
- 90 interaction rules for question generation
- Publicly available dataset (TREC-2007)

2.10 Paper 10: "Automatic question generation from text - an aid to independent study"

Authors: John H. Wolfe
Publication: United States Navy (SIGCUE Outlook, 1976)

What the Paper Says

This seminal paper describes AUTOQUEST, one of the earliest automatic question generation systems for educational purposes. The system uses pattern matching to generate questions from text paragraphs, presents them to students, and evaluates answers based on keyword matching. It pioneered syntactic pattern-matching approaches that influenced decades of QG research.

Technology/Models Used

- Pattern-matching algorithm (QGENR)
- Vocabulary dictionary (1700 verb stems)
- Morphological analysis for verb endings (-ing, -s, -ed, -es)
- Woods' Augmented Transition Network Parser (initial attempt, later abandoned)
- Sentence pattern tables (14 patterns in Table 1)

Detailed Steps for Model Training

Step 1: Preprocessing (QGENR preprocessor)

1. Check for exit commands (Goodbye, Quit)
2. Reject sentences with colons
3. Reject sentences >35 words
4. Strip leading filler words (nevertheless, however)
5. Mark -ly words as adverbs
6. Apply SETIONU to classify unknown words
7. Reject if first word is verb
8. Handle coordinated verbs (remove up to "and")
9. Apply CHECKVERBS to identify main verbs

Step 2: CHECKVERBS Algorithm

1. If verb preceded by determiner → mark as noun
2. If verb precedes another verb (first not copula) → mark first as noun
3. Verb after auxiliary → retain as verb

4. Verb after “to” → retain as verb
5. Verb after “be”/“have” → retain as verb
6. Count remaining verbs → compare with (relatives + subordinate conjunctions + 1)
7. If excess verbs, mark present tense verbs as nouns (right to left)

Step 3: Pattern Matching (14 patterns)
Example Pattern A (Subordinate conjunction):

```
Pattern: Prep (Subord + Z +, + X + Verb Aux + Y or X + Verb Aux + Y +
    Subord + Z

Case 1: X = nil or You + X and Tense = present
  Question: What should you do + Subord + Z?
  Answer: Verb + Y Aux

Case 2: Verb is intransitive
  Question: What happens with + X + Subord + Z?
  Answer: Verb + Y
```

Step 4: QFORM (Subject-Verb Inversion)

```
Rule 1: If conjunctions -> QFORM of first part + rest
Rule 2: If auxiliary -> move to front
Rule 3: If present tense verb -> replace with infinitive + "does" at front
Rule 4: If present tense verb -> "do" at front
Rule 5: If past tense verb -> "did" at front
```

Example from Paper:

```
Sentence: "The dd name identifies a DD statement so that subsequent control
     statements and the data control block in the processing program can
    refer to it."

Pattern match: "S1 so that S2" (Pattern C)

Step 1: Extract main verb "identifies"
Step 2: Create auxiliary "does" (present tense)
Step 3: Transform S1: "does the dd name identify"
Step 4: Question: "Why does the dd name identify a DD statement?"
Step 5: Answer: "So that subsequent control statements and the data control
     block in the processing program can refer to it."
```

Step 5: Answer Evaluation

1. Extract long words (>4 letters) from expected answer
2. Count matches in student’s answer
3. If >50% match → correct
4. If conjunction (and/or) → check each part separately
5. If partly correct → feedback: “Your answer is partly right.”

Evaluation Parameters

- Manual judgment of question quality
- Classification of errors into categories

Accuracy Results

Table 27: AUTOQUEST Accuracy Results

Result Category	Count	Percentage
Satisfactory questions	34	68%
Unsatisfactory questions	16	32%
- Syntactic errors	11	22%
- Semantic errors	3	6%
- Pedagogical errors	1	2%
- Program bugs	1	2%

Syntactic Errors Examples (Table 2):

1. "What are two usual characterizations..." → correct (good)
2. "What kind of proofs using each one?" → syntactic error
3. "What May 1974?" → syntactic error
4. "What happens with methods when a supposed match is invalid?" → embedding error

Semantic Errors Examples (Table 3):

1. "How soon is the primary goal of comprehension always to find meanings?" (abstract vs concrete)
2. "What does it assemble although FACT uses substantially more main memory?" (incorrect focus)
3. "What is the program reproduced in?" (idiomatic misinterpretation)

Publication Venue

SIGCUE Outlook, Volume 10, Issue SI, Pages 104-112 (1976) Association for Computing Machinery (ACM)

Key Contributions

- First systematic AQG system (AUTOQUEST)
- Pattern-matching approach for question generation
- Subject-verb inversion algorithm (QFORM)
- Verb identification algorithm (CHECKVERBS)
- 68% satisfactory question rate with 1970s technology

2.11 Paper 11: "Automatic question generation and answer assessment: a survey"

Authors: Bidyut Das, Mukta Majumder, Santanu Phadikar, Arif Ahmed Sekh
Publication: Research and Practice in Technology Enhanced Learning (2021)

What the Paper Says

This comprehensive survey covers automatic question generation and answer assessment strategies from textual and pictorial learning resources. It analyzes 78 articles from 2010-2020, categorizing approaches by question type (objective/subjective) and modality (text/image/audio-video). The authors identify challenges in informative sentence extraction, multi-sentence questions, answer assessment standards, and video-based assessment.

Technology Classification by Question Type

Objective Question Generation:

Table 28: Objective Question Generation Techniques

Question Type	Techniques	Key Papers
Open-cloze	Hint-based, word frequency	Pino & Eskenazi (2009), Agarwal (2012),
Cloze/MCQ	WordNet, pattern matching, ML	Coniam (1997), Brown et al. (2005), Mitk
Grammar tests	Handcraft patterns	Chen et al. (2006)
Vocabulary questions	WordNet, corpus analysis	Hoshino & Nakagawa (2007)

Subjective Question Generation:

Table 29: Subjective Question Generation Approaches

Approach	Method	Challenge
Taxonomy-based	Pre-coded taxonomy	Limited coverage
NLP + Bloom's	Dynamic generation	Memory optimization
Citation-based	Extract from academic writing	Domain specificity

Answer Assessment Techniques:

Table 30: Answer Assessment Techniques

Method	Technique	Accuracy
C-rater	Paraphrase recognition	84%
KNN + TF-IDF	Text categorization	76%
LSA/PLSA/LDA	Dimension reduction	Superior to KNN
M-BLEU	Modified BLEU	Higher than LSA
GLSA	Word n-gram inclusion	Better than LSA
Word ordering graphs	Pattern recognition	Effective with rubrics
Deep learning (LSTM)	Sequence analysis	Promising

Datasets Used

Dataset	Description	Application	Link
QGSTEC	1000 questions	QG evaluation	http://www.cs.memphis.edu/~vrus2/QGSTEC/
SQuAD	100K+ QA pairs	Reading comprehension	https://rajpurkar.github.io/SQuAD-explorer/
30MQA	30M QA pairs	QA training	Not publicly available
MS MARCO	1M questions	Machine reading	https://microsoft.github.io/msmarco/
RACE	97K MCQs	Reading comprehension	http://www.cs.cmu.edu/~glai1/data/race/
NewsQA	100K QA pairs	Machine comprehension	https://www.microsoft.com/en-us/research/project/newsqa-dataset/
TriviaQA	650K triples	Reading comprehension	http://nlp.cs.washington.edu/triviaqa/
NarrativeQA	46K QA pairs	Reading comprehension	https://github.com/deepmind/narrativeqa
TabMCQ	Crowdsourced MCQs	Multiple tasks	Custom
SciQ	13.7K MCQs	Science QG	https://allenai.org/data/sciq
MCQL	7.1K MCQs	Distractor generation	Custom
LearningQ	230K+ pairs	Educational QG	https://github.com/persist-gt/learningq

Visual Question Generation Datasets

Table 32: Visual Question Generation Datasets

Dataset	Description	Tasks
CLEVR	3D shapes	Visual reasoning
NLVR	2D shapes	Shape, size, color reasoning
VQA	Real-world images	Objects, color, counting
IQA	Events and actions	Interactive environments

Detailed Example of Open-Cloze Question Generation

Step 1: Sentence Selection

```
Select informative sentences from cricket news articles
Criteria: contains factual information about matches, players, scores
```

Step 2: Key Selection

```
Identify keywords (nouns, named entities)
Example key: "Sachin Tendulkar"
```

Step 3: Hint Generation

```
Hint level based on number of letters revealed
Level 1: first letter -> "S______"
Level 2: first 2 letters -> "Sa______"
Level 3: first 3 letters -> "Sac______"
```

Step 4: Question Formation

```
Original: "Sachin Tendulkar scored 100 runs in the match."
Open-cloze: "______ scored 100 runs in the match." (with hint: "S______")
```

Step 5: Evaluation Scoring

```
Score = f(number of hints used)
More hints used = lower score (indicates less knowledge)
```

Evaluation Parameters

For Question Generation:

- Well-formedness
- Relevance
- Grammatical correctness
- Difficulty level
- Educational usefulness

For Answer Assessment:

- Accuracy (vs human grading)
- Correlation with human scores
- Precision/Recall
- F-measure
- Cohen's kappa (inter-rater)

Accuracy Results

Publication Venue

Research and Practice in Technology Enhanced Learning (2021) 16:5 Springer Open

Table 33: Answer Assessment Accuracy Results

Study	**Method**	**Accuracy**
Leacock & Chodorow (2003)	C-rater	84%
Bin et al. (2008)	KNN + TF-IDF	76%
Dhokrat et al. (2012)	Taxonomy-based	70%
Sakaguchi et al. (2015)	Feature integration	Effective with limited data

Key Contributions

- Comprehensive review of QG and assessment
- Analysis of 78 articles (2010-2020)
- Identifies 7 major challenges:
  1. Informative-simple-sentence extraction
  2. Question generation from multiple sentences
  3. Short and long-type answer assessment
  4. Answer assessment standard
  5. Question generation from video lectures
  6. Assessment using machine learning
  7. Visual question generation

2.12 Paper 12: "Linguistic Considerations in Automatic Question Generation"

Authors: Karen Mazidi, Rodney D. Nielsen
Publication: Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (2014)

What the Paper Says

This paper presents a semantic role labeling-based automatic question generation system that uses pattern recognition to create questions of varying depth from expository text. The system achieves a 44% reduction in error rate compared to prior systems by leveraging semantic argument structures and careful pattern design. The authors also discuss linguistic challenges including negation detection, coreference resolution, and verb form handling.

Technology/ML Models Used

- **SENNA** (Collobert et al., 2011) for:
  - Tokenization
  - POS tagging
  - Syntactic constituency parsing
  - Semantic role labeling (PropBank 2005 scheme)
- **WordNet** (via NLTK) for verb lexical forms
- **Pattern matcher** with 42 patterns
- **Filter conditions** for question quality control

Dataset Used

- Science textbooks from www.ck12.org (open educational resources)
- 3 passages from Biology, Chemistry, Earth Science
- Average 60 sentences per passage (180 total sentences)
- Grade level: approximately 10th grade (readable-com)
- **Dataset link:** https://www.ck12.org/ (freely available)

Detailed Steps for Model Training

Step 1: Text Processing with SENNA

```
Input sentence -> SENNA pipeline:
- Tokenization: ["As", "this", "occurs", ",", "potential", "energy", "is",
    "released", "because", "the", "new", "bonds", "have", "lower", "
    potential", "energy", "than", "the", "original", "bonds", "."]
- POS tagging: [IN, DT, VBZ, ,, JJ, NN, VBZ, VBN, IN, DT, JJ, NNS, VBP, JJR
    , JJ, NN, IN, DT, JJ, NNS, .]
```

```
- Constituency parse: (S (SBAR (IN As) (S (NP (DT this)) (VP (VBZ occurs))) (, ,) (NP (JJ potential) (NN energy)) (VP (VBZ is) (VP (VBN released) (SBAR (IN because) (S (NP (DT the) (JJ new) (NNS bonds)) (VP (VBP have) (NP (JJ lower) (JJ potential) (NN energy)) (PP (IN than) (NP (DT the) (JJ original) (NNS bonds))))))) (. .))
- Semantic roles: For predicate "released.01":
    - A1: "potential energy"
    - AM-CAU: "because the new bonds have lower potential energy than the original bonds"
```

Step 2: Pattern Matching

```
For each predicate-argument set -> call matcher function
Pattern specification includes:
- Required arguments (e.g., A0, A1, ArgM)
- Filter conditions (e.g., !A0, !A1, !negation)
- Verb constraints (V=mean, !be)

Example pattern for "Why" questions:
Required: A1, ArgM-CAU
Filters: !A0, !A2
```

Step 3: Question Generation

From sentence: "As this occurs, potential energy is released because the new bonds have lower potential energy than the original bonds."

```
Match: Pattern requiring A1 + causation ArgM
- A1 = "potential energy"
- ArgM-CAU = "because the new bonds have lower potential energy than the original bonds"
Filters passed: no A0, no A2

Generated: "Why did potential energy release?"
Answer: "because the new bonds have lower potential energy than the original bonds"
```

Step 4: Verb Form Handling

```
Main verb "released" -> lexical form "release"
Tense: past (was released) -> auxiliary "did"
Subject-auxiliary inversion: "did potential energy release"
```

Step 5: Filter Application

Table 34: Filter Conditions for Question Quality

Field	Meaning
Ax	Must contain argument
!Ax	Must NOT contain argument
AxPER	Argument must refer to person
AxGER	Argument must contain gerund
AxNN	Argument must contain nouns
!AxIN	Argument cannot start with preposition
!AxPRP	Argument cannot start with personal pronoun
V=verb	Verb must be specific form
!be	Verb cannot be "be"
negation	Sentence cannot contain negation

Evaluation Parameters

- **Grammaticality** (1-5 Likert scale)
- **Clarity** (1-5 Likert scale)
- **Naturalness** (1-5 Likert scale)
- Inter-annotator agreement: 88% (within 1 point), Pearson's r = 0.47 ($p<0.001$)

Accuracy Results

Predicate Depth and Question Quality:

Table 35: Predicate Depth and Question Quality

Predicate Position	Questions	Average Rating
First	58	4.7
Second	35	4.7
Third	23	4.5
Higher	26	4.6

Comparison with Heilman & Smith (Chemistry: Bonds, 59 sentences):

Table 36: Comparison with Heilman & Smith

System	Grammaticality	Clarity	Naturalness	Average
H&S	4.38	4.13	3.94	4.15
M&N	4.76	4.26	4.53	4.52
Error Reduction	61%	15%	56%	**44%**

Comparison with Lindberg et al. (Earth Science: weather fronts, 93 sentences):

Table 37: Comparison with Lindberg et al.

System	Grammaticality	Clarity	Naturalness	Average
LPN&W	4.57	4.56	4.55	4.57
M&N	4.80	4.69	4.78	4.76
Error Reduction	54%	30%	51%	**44%**

Linguistic Challenges Identified

1. Negation Detection

```
Problem: "In Darwin's time and today, many people incorrectly believe that
    evolution means humans come from monkeys."
Negation in "incorrectly" not detected -> generates "What does evolution
    mean?" (should be avoided)
```

2. Coreference Resolution

```
Problem: "Air cools when it comes into contact with a cold surface or when it rises."
No coreference resolution -> generates "What happens when it comes into contact with a cold surface?"
Vague question with unresolved "it"
```

3. **Verb Forms**

- Light verbs: “make a decision”, “have a drink”, “take a walk”
- Catenative constructions: “began to cool”
- POS tagging errors: “use” classified as NN instead of VB

Publication Venue

Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (ACL 2014), pages 321-326

Key Contributions

- 44% error reduction over state-of-the-art
- Semantic role-based pattern matching
- 42 question patterns with filter conditions
- Detailed analysis of linguistic challenges
- Open-domain STEM textbook evaluation

2.13 Paper 13: "Literature Review of Automatic Question Generation Systems"

Authors: Sheetal Rakangor, Dr. Y. R. Ghodasara
Publication: International Journal of Scientific and Research Publications (2015)

What the Paper Says

This review paper analyzes 12 automatic question generation systems, comparing their algorithms, methodologies, question types, languages, and evaluation results. The authors focus on MCQ generation using NLP techniques including Named Entity Recognition (NER) and Semantic Role Labeling (SRL). They conclude that AQG remains an open research area with scope for generating additional question types (one-word answer, true/false).

Technology/ML Models Analyzed

System	Technology	Methodology
Cloze Question Generation	NER, domain-specific selection	Sentence selection → key selection → distractor selection
Discourse Connectives QG	Content selection + formation	7 discourse connectives (because, since, although, etc.)
Software Agents QG	Multi-agent system, Bloom's taxonomy	Document processing → classification → template-based
G-Ask	NER, citation classification	Citation extraction → classification → template generation
Automatic MCQ Generation	SRL, NER, similarity measures	Sentence extraction → similarity ranking → distractor selection
Semantic-based AQG	SRL, NER, Artificial Immune System	Feature extraction → pattern classification → question type testing
Multimedia-based QG	NLP, image matching	Video text extraction → human/system comparison → image selection

Predefined Templates QG	CWOQ algorithm	Knowledge descriptor → question generator → e-learning executer
Domain Ontology QG	OWL, ontology strategies	Class-based, property-based, terminology-based strategies
Punjabi QG	NER, rule-based	Name extraction → "who", location → "where", date → "when"
Gap Selection QG	Multiple features	Sentence selection → construction → classification/scoring
Linguistic Considerations QG	SENNA, SRL	Predicate-argument matching → pattern generation

Detailed Steps for Selected Systems

1. Cloze Question Generation (Narendra et al., 2013)

1. **Step 1: Sentence Selection**
   - Use MEAD summarizer to select informative sentences
   - Criteria: position, length, similarity to title
2. **Step 2: Key Selection**
   - Apply NER to identify named entities
   - Select keywords (nouns, named entities) as potential gaps
3. **Step 3: Distractor Selection**
   - Domain-specific (cricket articles)
   - Use web search and knowledge base for similar terms
   - Generate distractors based on semantic similarity

2. Automatic MCQ Generation (Fattoh, 2014)

1. **Step 1:** Extract sentences from dataset (145 parsed sentences)
2. **Step 2:** Prepare question sentence
   - Use SRL and NER to identify semantic patterns
   - Determine question type (who, what, where, when)

3. **Step 3:** Measure similarity
   - Compare question sentence with all sentences in knowledge base
   - Use string similarity measures
4. **Step 4:** Select distractors
   - Return three sentences with highest similarity
   - Extract keywords from these sentences as distractors

3. Semantic-based AQG (Fattoh, 2014)

1. **Step 1:** Input sentence processing
   - Apply SRL and NER for feature extraction
   - Convert to semantic pattern vectors
2. **Step 2:** Artificial Immune System
   - Feature extraction
   - Learning and storage in memory
   - Associative retrieval
3. **Step 3:** Pattern classification
   - Match sentence patterns to question type patterns
   - Training set: 250 patterns from 170 sentences
   - Question types: who, when, where, why, how
4. **Step 4:** Question generation based on classified pattern

Datasets Used

Table 39: Datasets Used in Analyzed Systems

System	Dataset	Size
Semantic-based AQG	Wikipedia/TREC 2007	170 sentences → 250 patterns
MCQ Generation	Custom	145 parsed sentences
Multimedia QG	Scientific video documentary	Custom
Predefined Templates	General Biology course (Bio110)	12 chapters, 239 subtopics, 46,345 questio
G-Ask	Student literature reviews	1,088 sentences, 221 citations

Evaluation Parameters

Accuracy Results

Publication Venue

International Journal of Scientific and Research Publications (IJSRP), Volume 5, Issue 1, January 2015

Table 40: Evaluation Parameters by System

System	Evaluation Method	Metrics
Cloze QG	Manual evaluation	Sentence quality, key quality, distractor c
Discourse Connectives	Manual	Semantic/syntactic soundness (2 evaluato
G-Ask	Cohen's Kappa, Precision/Recall	Question quality, citation classification
Semantic AQG	Precision, Recall, F-measure	Pattern classification accuracy
Ontology-based QG	Three dimensions	Pedagogical quality, linguistic correctness

Table 41: Accuracy Results

System	Accuracy
Automatic MCQ Generation	109/145 good sentences (75%)
Semantic-based AQG	87% truly generated patterns
Predefined Templates	46,345 questions generated
G-Ask (citation extraction)	65.60% (145/221 citations)
G-Ask (semantic correctness)	60% average

Key Contributions

- Comparative analysis of 12 AQG systems
- Detailed methodology descriptions
- Identifies research gaps: one-word answer, true/false questions
- Highlights importance of NER and SRL in AQG

2.14 Paper 14 & 15: "Automatic Question Generation: A Systematic Review"

Authors: Sonam Soni, Praveen Kumar, Amal Saha
Publication: SSRN Electronic Journal (2019)

What the Paper Says

This systematic review categorizes automatic question generation approaches into syntax-based, semantic-based, and template-based methods. The authors analyze 14 different algorithms with their procedures and question types, and discuss the role of Bloom's Taxonomy in question generation. They identify future research directions including true/false, analytical, and experimental question types.

Technology Classification by Approach

1. Syntax-based Approach (Most popular for factual questions)

- Uses Context Free Grammar parser
- Tregex for syntactic element identification
- Subject-auxiliary inversion for question formation
- Generates wh-movements

2. Semantic-based Approach

- Uses semantic parse (deeper than syntactic)
- Converts declarative semantic parse to questions
- Focuses on factoid-based questions
- Limited question quality

3. Template-based Approach

- No transformation rules needed
- Extracts informational content
- Uses question templates with slot fillers
- Generates deeper but less specific questions

4. Combined Syntax + Template Approach

- Better quality questions
- Uses original sentences with predefined templates
- Balances depth and specificity

Table 42: Bloom's Taxonomy Levels for Question Generation

Level	Cognitive Skill	Question Starters
1	Remembering	List, Define, Identify, Name
2	Understanding	Interpret, Summarize, Infer, Compare
3	Applying	Use, Execute, Implement, Show
4	Analyzing	Analyze, Explore, Infer, Diagram
5	Evaluating	Test, Judge, Reflect, Predict
6	Creating	Design, Combine, Create, Invent

Bloom's Taxonomy Integration

Detailed Algorithm Analysis

Sr No	Algorithm Name	Procedure	Question Types
1	Passage-based QG (Mazidi, 2017)	Topic modeling, noun phrase extraction, terminology extraction, heading analysis, NLG	Define, Describe, Give example, long descriptive
2	NLU to NLG (Mazidi & Tarau, 2016)	Semantic content analysis, NLU infused into NLG, template matching	Define, Describe, What
3	Factual Question Agent	Knowledge assessment, comprehension verification	Factoid
4	Gap-fill QG (Agarwal & Mannem, 2011)	Informative sentence extraction, key blanking, distractor selection	Gap-fill, MCQ
5	Cloze QG (Narendra et al., 2013)	Sentence selection, key selection, distractor selection (NER-based)	Cloze
6	Discourse Connectives (Agarwal et al., 2011)	Content selection, question formation (7 connectives)	Why, when, where, in which
7	Software Agents (Pandey & Rajeswari, 2013)	Document processing, information classification, template generation	Define, Describe, Give example
8	G-Ask (Liu et al., 2012)	Citation extraction, citation classification	Why, when, Does any

9	MCQ Generation (Fattoh, 2014)	Sentence extraction, similarity measurement, keyword selection	MCQ
10	Semantic-based (Fattoh, 2014)	Feature extraction (SRL, NER), pattern testing	Who, when, where, why, how
11	Multimedia-based (Skalban et al., 2012)	Text + image pre-questions, post-question analysis	Who, whom, where, whose, when, what
12	Predefined Templates (Awad & Dahab, 2014)	Knowledge descriptor, question generator, e-learning executor	OnlyOneSolution, TwoSolutions, AllOfTheAbove
13	Deep Reinforcement Learning (Kumar et al., 2018)	Seq2Seq + copy mechanism + coverage mechanism, RL loss + cross-entropy loss	Who, whom, where, whose, when, what
14	RNN for Factoids (Serban et al., 2016)	Neural network on knowledge base, text-to-question conversion	Factoid

Detailed Training Steps (Deep Reinforcement Learning Model)

Step 1: Encoder

$$\text{Input sequence } X = (x_1, x_2, \ldots, x_n)$$

Bi-LSTM encoder produces hidden states $h_1, h_2, \ldots, h_n$

$$h_t = [\text{forward } h_t;\ \text{backward } h_t]$$

Step 2: Decoder with Copy Mechanism

$$\text{At each time step } t:$$

- Compute attention over encoder states
- Generate probability distribution over vocabulary
- Copy probability for out-of-vocabulary words

$$p(y_t) = p_{\text{gen}} \times p_{\text{vocab}} + (1 - p_{\text{gen}}) \times p_{\text{copy}}$$

Step 3: Coverage Mechanism

$$\text{Coverage vector } c_t = \sum \text{attention weights up to } t$$

$$\text{Coverage loss} = \sum \min(\text{attention}_t, c_t)$$

Prevents word repetition

Step 4: Loss Function

$$\text{Combined loss} = -\sum \log P(y_t | y_{<t}, x) + \lambda \times \text{coverage_loss}$$

Reinforcement learning fine-tuning with task-specific rewards

Datasets Used

- **30M Factoid QA Corpus**: Freebase triples converted to QA (30M QA pairs)
- Custom datasets (varies by study)

Evaluation Parameters

- For RNN-based generation: Machine translation metrics, sentence similarity matrices
- For classification tasks: Precision, Recall, F-measurement
- For question quality: Human evaluation, automatic metrics

Accuracy Results

Table 44: Accuracy Results

Algorithm	**Accuracy/Performance**
Semantic-based AQG	87% truly generated patterns
RNN for Factoids	30M QA pairs generated
G-Ask	65.6% citation extraction, 60% semantic correctness

Publication Venue

SSRN Electronic Journal (2019) - https://ssrn.com/abstract=3403926

Key Contributions

- Systematic categorization of AQG approaches
- 14-algorithm comparative analysis
- Bloom's Taxonomy integration framework
- Deep reinforcement learning for QG
- Identifies future directions: true/false, analytical, experimental questions

3 Comprehensive Summary Table

Sr	Year	Authors	Title	Dataset Used	ML Model/Technology	Evaluation Parameters	Accuracy
1	2021	Kanchan Dhomse, Dr. Ishan Ranjan	Automatic Question Paper Generation using Bloom's Taxonomy Algorithm	Custom institutional database	Bloom's Taxonomy, Colony Algorithm, ML, NLP, Data Mining	Year-wise accuracy comparison (2016-2019)	ML&NLP: 95%, AI: 89%, Ontology: 86%
2	2022	Jawad Alshboul, Erika Baksa-Varga	A Review of Automatic Question Generation in Teaching Programming	Programming exercise repositories, custom collections	Bayesian Network, ITSB, ANN, Vortex Optimization, JUnit, ITA, Code-Training	User feedback, student performance, code coverage	Pre-COVID: 70-80%, Post-COVID: 85-90%
3	2021	Ruqing Zhang, Jiafeng Guo, Lu Chen, Yixing Fan, Xueqi Cheng	A Review on Question Generation from Natural Language Text	SQuAD, NewsQA, SearchQA, HotpotQA, MS MARCO, NarrativeQA, CoQA, QuAC, RACE, LearningQ	Seq2Seq (RNN/LSTM), Pre-trained (BERT, UNILM, ERNIE-GEN), Graph-based (GCN), VAE, GAN	BLEU-4, METEOR, ROUGE-L, Human evaluation	Best BLEU-4: 25.40 (ERNIE-GEN), Best ROUGE-L: 53.25

4	2020	Ghader Kurdi, Jared Leo, Bijan Parsia, Uli Sattler, Salam Al-Emari	A Systematic Review of Automatic Question Generation for Educational Purposes	SQuAD, NewsQA, SciQ, MCQL, QGSTEC, Medical CBQ	Syntactic-based, Semantic-based, Template-based, Rule-based, Statistical/ML	Acceptability, grammaticality, relevance, difficulty, distractor quality	66-94% sentence selection, 83-94% key selection, 60-88% distractors
5	2014	Ming Liu, Rafael A. Calvo, Vasile Rus	Automatic Generation and Ranking of Questions for Critical Review	504 questions from ACL/IJCAI papers, 489 from student reviews	RankSVM, Logistic Regression, LBJ NER, Tregex, Naive Bayes	Accuracy at top K, Cohen's Kappa, 5 quality measures	Top 25%: 75.8% (RankSVM), Top 50%: 71.3%
6	2020	Dhawaleswar Rao CH, Sujan Kumar Saha	Automatic Multiple Choice Question Generation From Text: A Survey	SQuAD, NewsQA, SciQ, MCQL, QGSTEC	NLP (POS, NER, parsing, SRL), WordNet, Ontology, ML classifiers	Well-formedness, difficulty, relevance, grammaticality, distractor plausibility	66-94% sentence selection, 79-94% key selection, 60-88% distractors
7	2017	Manisha Divate, Ambuja Salgaonkar	Automatic question generation approaches and evaluation techniques	Multiple (varies by study)	Template-based, Syntax-based, Semantic-based, Logistic Regression, RankSVM, J48	Precision, Recall, F-score, Cohen's Kappa, syntactic deficiency analysis	Precision: 0.47-0.79, Recall: 0.27-0.71, F-score: 0.38-0.70

8	2010	Ming Liu, Rafael A. Calvo, Vasile Rus	Automatic Question Generation for Literature Review Writing Support	6 literature reviews (1,088 sentences, 221 citations)	LBJ NER, Tregex, Stanford Parser, WordNet, SENTIWORDNET, Rule-based classification	Citation extraction (65.6%), semantic correctness (60%), 5 quality measures, Bystander Turing Test	60% semantic correctness, F-score 0.43 in Turing test
9	2010	Husam Ali, Yllias Chali, Sadid A. Hasan	Automatic Question Generation from Sentences	TREC-2007 QA Track (70 topics, 50 files each)	Oak system, Charniak parser, Two-layered classifier, 90 interaction rules	Recall, Precision	Recall: 27.6%, Precision: 58.7%
10	1976	John H. Wolfe	Automatic question generation from text - an aid to independent study	Technical report abstracts, IBM programmer's manual	Pattern-matching (QGENR), CHECK-VERBS, QFORM, Woods' ATN parser	Manual error classification	68% satisfactory, 32% unsatisfactory

11	2021	Bidyut Das, Mukta Majumder, Santanu Phadikar, Arif Ahmed Sekh	Automatic question generation and answer assessment: a survey	QGSTEC, SQuAD, 30MQA, MS MARCO, RACE, NewsQA, TriviaQA, NarrativeQA, TabMCQ, SciQ, MCQL, LearningQ, CLEVR, NLVR, VQA, IQA	C-rater, KNN+TF-IDF, LSA/-PLSA/LDA, M-BLEU, GLSA, LSTM, Word2vec, Seq2Seq	Accuracy, correlation, precision, recall, F-measure, Cohen's kappa	C-rater: 84%, KNN: 76%, Taxonomy-based: 70%
12	2014	Karen Mazidi, Rodney D. Nielsen	Linguistic Considerations in Automatic Question Generation	ck12.org science textbooks (Biology, Chemistry, Earth Science)	SENNA, WordNet, Pattern matcher (42 patterns)	Grammaticality, clarity, naturalness (1-5 Likert), inter-annotator agreement	44% error reduction, ratings: 4.52-4.76
13	2015	Sheetal Rakangor, Dr. Y. R. Ghodasara	Literature Review of Automatic Question Generation Systems	Wikipedia, TREC 2007, General Biology course, student reviews	NER, SRL, Artificial Immune System, OWL, MEAD summarizer, CWOQ algorithm	Manual evaluation, semantic/syntactic soundness, precision/recall	75% good sentences, 87% patterns, 60% semantic correctness

14	2019	Sonam Soni, Praveen Kumar, Amal Saha	Automatic Question Generation: A Systematic Review	30M Factoid QA Corpus, custom datasets	Syntax-based, Semantic-based, Template-based, Deep Reinforcement Learning, RNN, Bloom's Taxonomy	Precision, Recall, F-measure, machine translation metrics	87% patterns, 30M QA pairs generated

4 Recommendations for Gujarati Question Answer Generation System

Based on the comprehensive literature review, here are the critical insights and recommendations for developing an Automatic Question Answer Generation system in Gujarati:

4.1 Language-Specific Challenges

- Most existing work focuses on English (59 studies), with limited work in Chinese (5), Japanese (3), Indonesian (2), Punjabi (1), Thai (1)
- Punjabi QG system (Paper 13) used NER and rule-based approach for "who/where/when" questions - similar approach adaptable for Gujarati
- Need for Gujarati-specific resources: NER tagger, POS tagger, parser, WordNet-like lexical database

4.2 Technology Selection Recommendations

Table 46: Technology Selection by Question Type

Question Type	**Recommended Approach**	**Key Papers**
Factoid (who/what/where/when)	NER + SRL + template-based	Papers 1, 2, 8, 9
MCQ	6-phase workflow + WordNet-like ontology	Paper 6
Gap-fill	Sentence selection + key selection + distractor generation	Papers 6, 11
Deep learning questions	Semantic patterns + higher-order templates	Papers 5, 7, 12

4.3 Dataset Requirements for Gujarati

- Need to create or adapt: Gujarati textbook corpus, news articles, Wikipedia content
- Parallel English-Gujarati corpus for transfer learning
- Question-answer pairs in Gujarati for training/evaluation (your JSON dataset is an excellent start)

4.4 Evaluation Metrics to Adopt

- BLEU, ROUGE, METEOR for automatic evaluation (Paper 3)
- Human evaluation: grammaticality, clarity, relevance, usefulness (Papers 5, 8, 12)
- Cohen's Kappa for inter-annotator agreement (Papers 5, 8)
- Precision/Recall for component evaluation (Papers 7, 9)

4.5 Training Approach for Gujarati

1. **Data Collection:** Gather Gujarati textbooks, articles, and manually create QA pairs (your JSON dataset provides this foundation)
2. **Resource Development:** Build/train Gujarati NER, POS tagger, parser using tools like iNLTK (Indian Natural Language Toolkit)
3. **Model Selection:**
   - Start with template-based approach (less data hungry) using your JSON dataset patterns
   - Progress to neural methods as data grows
   - Consider mBERT or IndicBERT for transfer learning
4. **Transfer Learning:** Leverage multilingual BERT/mBERT/IndicBERT fine-tuned on Gujarati
5. **Evaluation:** Adapt English evaluation metrics for Gujarati

4.6 Recommended ML/DL Models for Your Dataset

Based on your JSON dataset structure with multiple domains and question types, here are the recommended models:

Table 47: Recommended Models for Gujarati QG

Model Type	**Description**	**Justification**
Seq2Seq with Attention	Encoder-decoder with attention mechanism	Handles variable-length inputs, proven for QG (Paper 3)
Transformer-based (IndicBERT/mBERT)	Pre-trained multilingual transformer	Transfer learning for low-resource languages
T5 (Text-to-Text Transfer Transformer)	Unified framework for text generation	State-of-the-art for QG tasks
Rule-based + ML Hybrid	Template matching + ML ranking	Less data hungry, interpretable (Papers 5, 12)

4.7 Implementation Roadmap

1. **Phase 1:** Develop Gujarati NLP tools (tokenizer, POS tagger, NER) using iNLTK or train from scratch
2. **Phase 2:** Implement template-based QG using patterns from your JSON dataset
3. **Phase 3:** Train Seq2Seq model on your dataset (split 80-10-10 for train-val-test)
4. **Phase 4:** Fine-tune IndicBERT/mBERT on your Gujarati QA pairs
5. **Phase 5:** Evaluate using BLEU, ROUGE, and human evaluation

5 Conclusion

This comprehensive literature review has analyzed 16 seminal papers in the field of automatic question generation, spanning from 1976 to 2022. The review has covered traditional rule-based approaches, machine learning techniques, and modern deep learning methods, providing a solid foundation for developing a Gujarati question answer generation system.

The review has identified that while significant progress has been made in English and other major languages, work on Indic languages like Gujarati remains limited. Your JSON dataset, with its multi-domain context and diverse question types, provides an excellent starting point for this research.

Based on the findings, a hybrid approach combining template-based methods for common question types and deep learning (Seq2Seq/Transformer) for complex questions is recommended. The evaluation framework should incorporate both automatic metrics (BLEU, ROUGE) and human evaluation for linguistic quality.

Future work should focus on developing Gujarati-specific NLP resources, expanding the dataset, and exploring transfer learning from multilingual models.

Table 48: Summary of All Models with Paper References

#	Model Type	Model Name	Paper References
1	Classification	Naive Bayes	5, 8
2	Classification	Logistic Regression	5, 7
3	Classification	SVM	5, 6, 7
4	Ranking	RankSVM	5, 7
5	Classification	Decision Trees	7
6	Classification	J48	7
7	Classification	KNN	11
8	Regression	Linear Regression	3, 5, 7
9	Clustering	K-Means	3
10	Topic Modeling	LDA	3, 6, 11
11	Feature Extraction	TF-IDF	6, 11
12	Embeddings	Word2Vec	6, 11
13	Embeddings	FastText	11
14	NER	LBJ NER	5, 8
15	NER	Stanford NER	3, 6

#	**Model Type**	**Model Name**	**Paper References**
16	Sequence Labeling	CRF	3, 6
17	Sequence Labeling	HMM	2
18	Parsing	Stanford Parser	3, 5, 8
19	Parsing	Charniak Parser	7, 9
20	NLP Pipeline	SENNA	7, 12, 13
21	SRL	Illinois SRL	7, 9
22	SRL	PropBank	7, 12
23	Seq2Seq	Basic Seq2Seq	3, 11, 14
24	Seq2Seq	Seq2Seq + Attention	3, 14
25	Seq2Seq	Bi-LSTM	3, 14
26	Seq2Seq	Copy Mechanism	3, 14
27	Seq2Seq	Coverage Mechanism	3, 14
28	RNN	LSTM	3, 11, 14
29	RNN	GRU	3
30	RNN	Stacked LSTM	3
31	Transformer	BERT	3
32	Transformer	mBERT	3
33	Transformer	T5	3
34	Transformer	mT5	3
35	Transformer	BART	3
36	Transformer	ProphetNet	3
37	Transformer	UNILM	3
38	Transformer	ERNIE-GEN	3
39	Indic	IndicBERT	14
40	Indic	IndicBART	14
41	Indic	iNLTK	14
42	Generative	VAE	3, 14
43	Generative	CVAE	3
44	Generative	GAN	3, 14
45	Optimization	Vortex Algorithm	2
46	Optimization	Colony Algorithm	1
47	Evaluation	BLEU	3, 7, 11
48	Evaluation	ROUGE	3, 7, 11
49	Evaluation	METEOR	3, 7
50	Evaluation	Cohen's Kappa	5, 7, 8
51	Hybrid	Rule + ML	1, 5, 12
52	Hybrid	Syntax + Template	14
53	Domain	Bloom's Classifier	1
54	Domain	Citation Classifier	5, 8
55	Domain	MCQ Generator	6
56	Domain	Distractor Generator	6
57	Domain	Sentence Simplifier	6, 7